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Beyond Linear Limits: In-House ML Soft Sensors Transforming APC with HPINSYT

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Abstract

Advanced Process Control (APC) systems depend on inferential modeling to predict product quality in near real time, particularly in scenarios where analyzer data is unavailable or lab test results are delayed. Historically, such inferential models have relied on linear regression, which often fails to capture the dynamic and non-linear behavior typical of modern refining processes. As refining operations grow more non-linear and responsive, such conventional models often fall short in providing accurate, timely estimates—leading to suboptimal control and operational inefficiencies.

To overcome these constraints, Hindustan Petroleum Corporation Limited (HPCL) developed a refinery-wide framework for machine learning (ML)-based soft sensors, aimed at replacing or augmenting traditional APC inferential. These soft sensors were built using supervised ML models trained on historical DCS and laboratory data, carefully validated to ensure first-principle consistency and explainability. The models predict critical quality parameters in near real time using streaming process data, ensuring high availability of quality estimates—even during analyzer downtimes or lab result delays.

Over 40 soft sensors were developed and deployed across major refinery units such as CDU, VDU, FCC, CCR, DHT, SRU etc. Importantly, a subset of these models was integrated into closed-loop APC systems, where soft sensor outputs now serve as active feedback for control decisions—demonstrating their stability, accuracy, and reliability in live operations.

The development and deployment lifecycle of these models was enabled by HPINSYT—HPCL's in-house data analytics platform—which provided a no-code interface for model building, evaluation, deployment, and simulation. This enabled close collaboration between domain experts and data scientists, dramatically accelerating development cycles and improving adoption on the plant floor.

This paper presents the methodology, model architecture, APC integration strategy, and field results from HPCL's ML-based soft sensor deployments, highlighting their role in elevating APC systems beyond linear limitations.

Introduction

Refineries depend on inferential models for real-time visibility into key product quality parameters that cannot be measured continuously. These include octane number, sulfur content, benzene levels, and

viscosity—each critical for product compliance, energy efficiency, and environmental control. Traditionally, these inferential are built into APC layers using **linear regression, PLS, WLS etc.**, which assumes fixed relationships between a set of process inputs and the quality variable.

While this approach works under narrow, steady-state conditions, it has several drawbacks:

- It fails to model **non-linear dependencies** common in refining processes (e.g., temperature-vs-sulfur curves).
- It struggles with **changing process regimes**, such as feedstock variability or catalyst aging.
- It depends on **periodic manual calibration** and cannot adapt to subtle drifts over time.

In parallel, **online analyzers** though more accurate, suffer from maintenance requirements, high capital cost, and frequent calibration needs, being high on CAPEX, analyzers maynot be installed for all kinds of quality analysis. Meanwhile, laboratory testing introduces **significant time lag**, making it unsuitable for real-time control.

These challenges motivated HPCL to focus on the **development of soft sensors**—machine learning models that use DCS data to predict quality variables in real time. Soft sensors offer a middle ground between lab measurements and analyzers: **low-latency, data-driven predictions with high coverage and low cost.**

The primary goals of the initiative were:

- To develop **robust, interpretable ML models** for quality prediction.
- To integrate these soft sensors into **live APC systems** for closed-loop control.
- To ensure **scalability across units**, with minimal customization effort.
- To enable **domain experts** to participate in development without needing programming skills.

HPINSYT was developed internally to facilitate these goals—but the central innovation lies in the soft sensor models themselves and their **operational integration** with APC frameworks.

System Architecture and Methodology

Selection of Use Cases

Soft sensor development began with identifying quality variables that were:

- Critical for **product quality assurance**
- **Unavailable in real time** (no analyzer or frequent lab delays)
- Already being **used or needed** within APC strategies

Initial deployments targeted:

- **Reformat and Isomerase RON**
- **Diesel Product and Feed Sulfur**
- **SRU Air Demand (O₂ Balance)**
- **Naphtha Benzene Content**
- **Lube Feedstock properties**
- **PSA Inlet Methane Content**

- **Blending component properties**

Each of these variables was previously estimated via delayed lab results or static inferentials—often unreliable during transients.

Data Preparation

Historical datasets spanning 6–48 months were extracted from the refinery’s DCS and LIMS systems. Each dataset underwent rigorous cleaning and pre-processing, including:

- Removal of flat-lined or outlier values (sensor faults)
- Time alignment of lab results with corresponding DCS snapshots
- Interpolation of missing values with process-aware methods
- Deadband filtering to reduce control noise artifacts

Model Development

HPCL used supervised machine learning models to correlate real-time DCS input tags with lab-measured quality parameters, selecting algorithms like XGBoost and LightGBM based on their ability to model non-linear dependencies:

- Gradient Boosted Decision Trees (LightGBM, XGBoost)
- Random Forests
- Support Vector Machine
- Linear Models (Linear, Lasso, Ridge)

Feature selection was guided by both domain knowledge and statistical analysis (correlation, permutation importance). Input tags typically included:

- Reactor temperatures and pressures
- Flowrates and valve positions
- Composition estimates from upstream analyzers

Each model was subjected to:

- **Cross-validation** to test generalization
- **Residual plots** to assess accuracy
- **Model Interpretability Plots (XAI)** for feature interpretability
- **Process sanity checks** by domain experts

Models were rejected if they violated known physical relationships (e.g., increasing reactor severity leading to higher sulfur), even if statistically accurate.

Deployment and APC Integration

Real-Time Inference Layer

Validated models were packaged and deployed as containerized services on HPCL's internal infrastructure and services, enabling real-time execution and integration with existing control systems. These services:

- Pull live DCS data at a predefined frequency
- Preprocess inputs (scaling, transformation)
- Run inference and produce predictions
- Send results to process dashboards and historians

The system includes error-handling, data buffering, and retry logic to ensure robust inference even during network or tag disruptions.

Integration with APC Systems

One of the defining outcomes of the soft sensor initiative was **closed-loop integration with APC**. The outputs of selected soft sensor models were incorporated into APC control loops, either replacing existing static inferentials or serving as parallel inputs for performance enhancement.

Examples include:

- **Reformat RON and Benzene**
- **Isomerization Unit Feed Quality and Product RON**
- **Lube Section Feed Stock and Product Properties**
- **FCCU LCN RON**

Key requirements for APC integration:

- High **prediction stability** (no sudden jumps)
- **Robustness to outliers**
- Consistent **latency** (fixed refresh rate)
- Acceptance testing with APC engineers under multiple scenarios

Results showed that soft sensors:

- Improved controller response time
- Reduced oscillations during feed changes
- Enabled operation closer to product specs with tighter margins
- Use cases which were earlier not feasible with APC inferentials were also possible using this approach

Role of HPINSYT in Soft Sensor Development

While machine learning-based soft sensors at HPCL were developed using both code-based and GUI-driven approaches, **HPINSYT** served as a powerful enabler by streamlining and standardizing the development process—especially for cross-functional collaboration. Its no-code architecture made the creation and

deployment of soft sensors more accessible and scalable across refinery units. Key roles played by HPINSYT include:

- **No-Code Enablement for Domain Experts**
Allows process engineers without programming backgrounds to contribute directly to model development—performing data exploration, feature selection, and model validation.
- **Streamlined Development Workflow**
Provided a unified interface for data ingestion, preprocessing, training, and evaluation—reducing development time and standardizing practices across units.
- **Complementary to Code-Based Models**
While some soft sensors were developed via scripting for complex customization, HPINSYT supported rapid prototyping, iteration, and deployment for a wide range of use cases.
- **Built-In Visualization and Explainability**
Included tools for trend analysis, model interpretability, and "what-if" simulations, enabling deeper insights and better operator adoption.
- **Facilitated Collaboration Across Roles**
Bridged the gap between data scientists and operations teams, fostering a more integrated model development and validation cycle.
- **Deployment Ready**
Enabled one-click deployment of validated models as real-time soft sensors, with support for integration into dashboards and APC interfaces.

HPINSYT served as the end-to-end platform for:

- **Data ingestion and cleansing** from DCS and LIMS sources
- **Exploratory data analysis** and variable selection
- **Model training, validation, and visualization**
- **Real-time deployment** of trained models for continuous prediction
- **‘What-if’ simulation and interpretability tools** for domain experts
- **Case-Study** – An deployed model can be used for carrying out sensitivity analysis and generation of delta-vectors.

In essence, **HPINSYT did not replace code-based modelling**, but significantly enhanced the speed, accessibility, and sustainability of soft sensor development—particularly where tight collaboration with domain experts was essential.

Results and Observations

Model Performance

The success of any machine learning-based soft sensor hinges on three fundamental criteria: accuracy, stability, and resilience to operating variability. Each model deployed in HPCL’s refinery-wide ML soft sensor initiative was evaluated against these criteria to ensure reliability and consistency under industrial operating conditions.

Performance Metrics

Across all units, the soft sensor models consistently achieved high levels of predictive accuracy. For each deployment, the following metrics were calculated and benchmarked against laboratory tolerance bands:

- **Root Mean Squared Error (RMSE):** Typically within 3–10% of the measured lab value, depending on the property.
- **Mean Absolute Error (MAE):** Maintained within $\pm$ lab repeatability tolerance for most models.
- **Coefficient of Determination (R^2):** Ranged from 0.60 to 0.98 across most use cases, indicating strong correlation between predicted and true values.
- **Bias (Mean Error):** Regularly monitored and minimized through retraining or calibration, especially when lab procedures evolved.

Example:

- **LCN RON Prediction (NFCCU):**
 - RMSE: 0.35 units
 - MAE: 0.3 units
 - R^2 : 0.8

Stability and Noise Immunity

Unlike online analyzers that can suffer from drift, maintenance noise, or lag during recalibration, the soft sensors showed excellent **temporal smoothness**, with low high-frequency noise. This made them ideal for APC integration, as controller stability depends on predictable, smooth inputs.

Process Unit Coverage

Soft sensors have been deployed across a diverse range of units, including:

- Crude and Vacuum Distillation Units (CDU/VDU)
- Fluid Catalytic Cracking (FCC)
- Continuous Catalytic Regeneration (Naptha Reforming – CCR)
- Isomerization
- Diesel Hydrotreating (DHT)
- Sulfur Recovery Units (SRU)
- Lube Operations
- Blending operations

This **unit-agnostic architecture** proves the framework's scalability and adaptability.

Benefits Realized

- Enhanced real-time quality visibility
- Improved APC performance due to more accurate inferentials
- Availability of Inferentials which were not Possible earlier due to non-linearity
- Reduced manual sampling delays and operator workload
- Higher operational agility during feed/product transitions

Discussion

Moving Beyond Linearity: Rethinking Inference in the ML Era

Traditional inference in Advanced Process Control (APC) systems have been rooted in the mathematical comfort of linear models—simple to implement, quick to calibrate, and computationally inexpensive. However, these very characteristics that once made them attractive have now become their greatest limitations. In the modern refinery landscape, characterized by increasing feedstock variability, dynamic operating strategies, aging catalysts, and more stringent product specifications, the assumptions of linearity are no longer tenable.

ML-powered soft sensors offer a solution that effectively addresses the challenges of non-linearity and variability in refinery operations, capturing dynamic behavior that traditional models cannot. Unlike linear regression models that assume constant slopes and intercepts, ML models—especially decision tree ensembles like XGBoost and LightGBM—can detect conditional patterns, sudden jumps, and saturation effects that occur in real-life refining processes. For example, the impact of reactor temperature on sulfur content may follow a logarithmic trend at lower severities but a diminishing-return regime at higher severities. Such behavior, while challenging to model linearly, is naturally captured by data-driven models.

Further, ML soft sensors can learn and incorporate cross-variable interactions that evolve over time. For instance, in FCC or CCR units, the impact of feed preheat temperature on final product RON may depend not only on the current flow rate but also on recent catalyst circulation patterns, which may have temporal memory effects. These non-obvious interactions are missed by conventional models, often requiring manual tuning or introduction of engineered features that are brittle or inconsistent.

Another notable strength of ML-based inference is their ability to remain robust under sudden operating regime shifts—such as crude switches, catalyst regeneration events, or process upsets. Static inference typically exhibit poor extrapolation under such transitions, whereas machine learning models, trained on rich historical data, are more adaptable and less sensitive to small input perturbations.

As a result, these soft sensors don't just offer better predictions—they offer predictions that remain useful across time, context, and varying operating philosophies. This reliability translates directly into better APC performance: tighter control bands, improved setpoint tracking, and reduced controller hunting.

Enabling Human-Machine Collaboration: The HPINSYT Paradigm

HPINSYT stands out by empowering process engineers, APC specialists, and control room personnel to directly participate in the ML model lifecycle—without writing a single line of code. Its no-code environment includes guided interfaces for dataset selection, tag filtering, exploratory data analysis, feature engineering, and model evaluation. By decoupling domain expertise from coding requirements, HPINSYT breaks down traditional silos that hinder industrial AI projects.

This democratization is not just a technical convenience—it is a strategic advantage. Domain experts bring decades of process knowledge, causal understanding, and physical intuition, which are invaluable during model training and validation. Their active involvement ensures that models are not just accurate, but meaningful, interpretable, and aligned with process logic.

For example, a model predicting SRU Air Demand (O_2 balance) may statistically correlate with vent valve position, but a process engineer would flag this as an effect rather than a cause, prompting refinement in feature selection. Such insights, when incorporated iteratively, yield models that are more robust and acceptable for integration.

Enhancing Trust through Explainability

One of the key barriers to adopting ML in process control has historically been its "black box" nature. Operators and engineers are understandably hesitant to rely on predictions they cannot interpret—especially when these influence critical control decisions.

HPINSYT addresses this challenge head-on by integrating explainability tools, sensitivity plots, and real-time trend overlays. These tools allow engineers to visualize not just what the model is predicting, but why. For example, a sudden drop in predicted RON can be traced back to upstream changes in feed composition or temperature excursions, enabling better root cause analysis.

This transparency has played a pivotal role in gaining operator confidence and ensuring the models are not just deployed, but actively used and trusted on the plant floor.

Customization, Security, and Ownership: Why In-house Matters

Off-the-shelf ML platforms often suffer from a one-size-fits-all approach, lack of transparency in algorithmic logic, and limited customization capabilities. For industrial applications like refineries—where data formats, system architectures, and security protocols vary widely—these limitations are significant.

HPINSYT, being internally developed and maintained, ensures:

- **Complete control over modeling logic:** Algorithms can be tweaked or extended to incorporate physical constraints, safety rules, or custom loss functions and future new requirements.
- **Secure integration with refinery networks:** No data leaves the refinery premises, eliminating concerns over cloud compliance or vendor lock-in.
- **Seamless compatibility with APC, DCS, and historian systems:** Because the platform is tailored for HPCL's infrastructure, deployment and integration timelines are drastically reduced.
- **Agility in updates and evolution:** As new use cases emerge; models and workflows can be rapidly adapted without vendor dependency.

This deep customization has also allowed HPINSYT to evolve into more than just a modeling platform. It is now being leveraged for what-if simulations, control impact assessments, and sensitivity analysis, providing a unified decision support environment.

Operational Resilience: Lessons from Field Deployments

Real-world deployments have uncovered several best practices that now inform future model development at HPCL:

- **Retraining Frequency:** Some models need quarterly updates due to drift in feed composition and gaussian mean of process data; others have remained stable for over 6-7 months.
- **Model Governance:** A model registry, validation logs, and approval workflows ensure traceability and audit-readiness.
- **Fallback Logic:** Soft sensors are designed with graceful degradation—if input tags are unavailable or noisy, the prediction is flagged or held to the last known safe value.
- **Impact Auditing:** Improvements in APC KPIs (like reduced oscillations, tighter margin control, or analyzer backup time) are continuously monitored.

In this broader context, ML-based inferentials are not just passive predictors—they are enablers of a more intelligent, resilient, and agile control architecture.

Conclusion

Summary

The deployment of ML-based soft sensors at HPCL, supported by the in-house HPINSYT platform, represents a transformative leap in the evolution of Advanced Process Control (APC) in refinery operations. Traditionally, APC inferentials have relied on linear models and simplistic assumptions that fail to reflect the complexities of modern refining. As units become more integrated, feedstocks more variable, and

operational targets more stringent, the need for intelligent, adaptable, and interpretable inferentials has become paramount. HPCL's initiative addresses this gap head-on.

The core innovation lies in enabling real-time, data-driven prediction of critical quality parameters using historical laboratory and DCS data. By doing so, ML soft sensors bridge the long-standing gap between the reliability of analyzers and the speed of lab results. They offer refinery engineers a powerful new instrument—one that enhances control accuracy, reduces lag in decision-making, and brings process insights into sharper focus.

But technical accuracy alone is not enough. The success of this initiative also stems from its human-centered approach. HPINSYT was deliberately designed to bring domain experts—process engineers, APC specialists, and operations teams—into the AI development loop. With its no-code modeling environment, built-in visualizations, and explainability features, HPINSYT democratized ML development across disciplines, breaking the traditional barriers between data science and plant operations.

Furthermore, the integration of soft sensors into live APC systems showcases not only the models' technical reliability but also their operational maturity. From FCCU to CCR and from SRU to Lube units, soft sensors have proven their utility in real-world closed-loop control environments. They have demonstrated tangible benefits—enhanced real-time visibility, reduced control oscillations, minimized analyzer downtime impact, and enabled tighter operation within quality margins. Use cases that were previously infeasible due to non-linearity or lack of analyzers are now fully operational with ML-based inferentials.

An important highlight of the project is the development of a scalable, reusable architecture. Rather than building one-off models, the approach was to create a standardized pipeline—from data ingestion to deployment—that can be applied across units and use cases with minimal rework. This ensures long-term sustainability, rapid replication, and easier governance of the ML models.

By investing in an in-house platform like HPINSYT, HPCL has ensured full control over model development, data security, and integration strategy. This also provides agility for future innovations—whether it be retraining models as processes evolve, embedding first-principle logic into ML pipelines, or extending soft sensors to predictive maintenance, emissions monitoring, or sustainability optimization.

In conclusion, the Transition to ML-based soft sensors represents a strategic evolution in how HPCL embeds process intelligence into control systems, moving beyond traditional linear models to more adaptive and insightful solutions. HPCL's refinery-wide soft sensor deployment demonstrates that with the right platform, the right people, and the right process, AI can deliver real, measurable, and sustainable value in core industrial operations.

Future Work

- **Integration with Maintenance and Inspection**

Soft sensors will be linked with mechanical integrity systems to predict potential equipment failures, enabling proactive maintenance and reducing unplanned shutdowns.

- **Model Drift Detection and Auto-Retraining**

Automated pipelines will monitor model performance over time, detect drift, and trigger retraining using fresh data to maintain accuracy and reliability.

- **Hybrid ML-Physics Model Development**

Combining machine learning with first-principal constraints will improve interpretability, safety compliance, and robustness across varying operating regimes.

- **Expansion to Emissions and Energy Optimization**

Soft sensors will be extended to forecast emissions and optimize energy use, supporting HPCL's sustainability and carbon reduction initiatives.

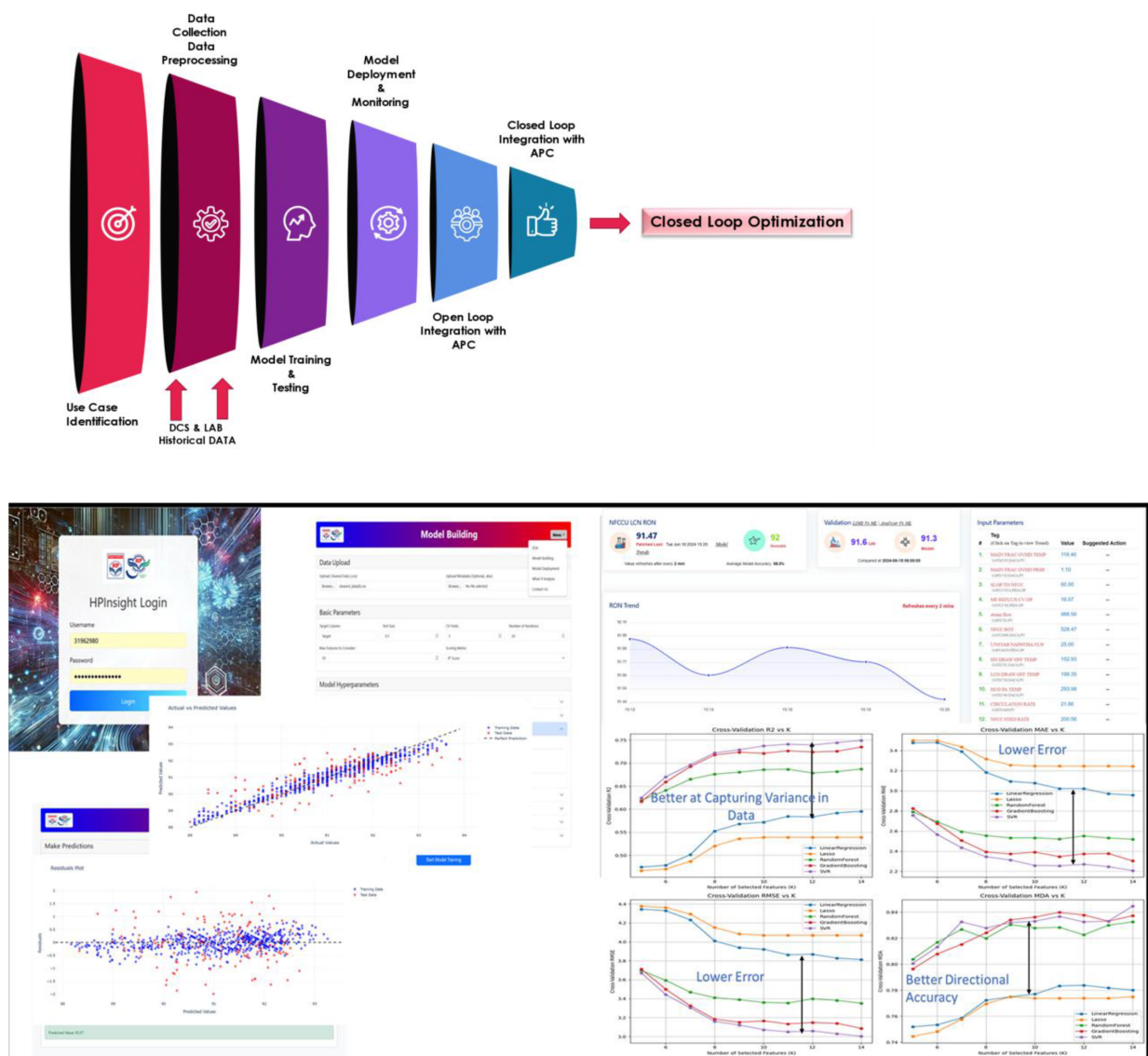


Figure 1—HPINSYT Soft Sensor Workflow: From lab data and DCS ingestion to EDA, Model Training, Comparison and Deployment

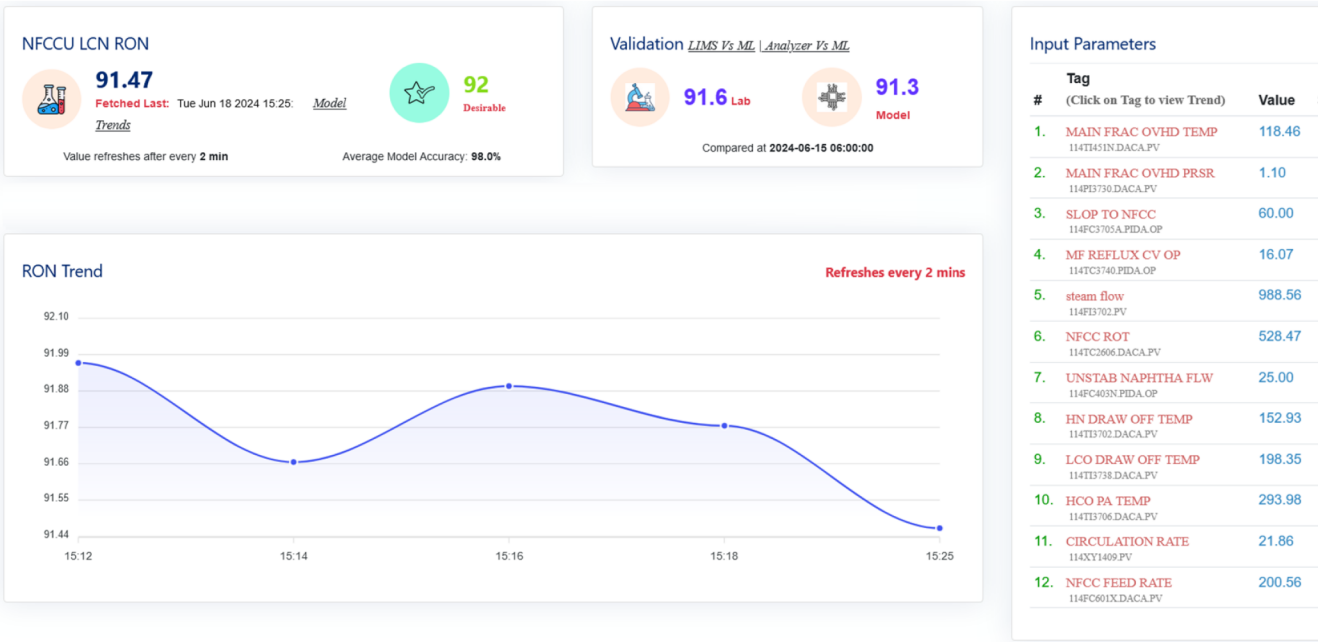


Figure 2—Real-Time Soft Sensor Dashboard showing minute-wise predictions and trends

